

Optimized Random Forest Classification for Enhanced Abnormality Detection in Pressurized Water Reactors

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Abstract— Ensuring safety and operational integrity in Nuclear Power Plants (NPPs) requires accurate and timely detection of sensor abnormalities within Pressurized Water Reactors (PWRs). Bias and drift faults in power, pressure, and vibration sensors can compromise monitoring accuracy, while the indirect nature of vibration signals further complicates diagnosis. This study proposes an optimized Random Forest (RF) classification framework for detecting such anomalies in PWRs. A publicly available dataset, consisting of 498 normal and 50 abnormal instances, was employed as a proof-of-concept. To address the inherent class imbalance, the Synthetic Minority Oversampling Technique was applied during preprocessing, and hyperparameters were tuned through Grid Search to enhance model performance. The optimized RF achieved a high accuracy of 96% with a precision of 0.98, recall of 0.93, F1-score of 0.96, and an AUC of 0.996 in identifying both bias and drift faults. With a modest training time of 13.4 seconds, the proposed framework highlights a pathway toward precise anomaly detection, contributing to improved fault prevention, reactor reliability, and safe power generation. While the limited dataset size presents potential risks of inflated performance due to oversampling, the results show the potential of this RF-based method as a proactive diagnostic tool for PWRs. Future work will focus on validation with larger and more representative PWR datasets to ensure robustness and generalizability.

Keywords— *Pressurized Water Reactor, Abnormality Detection, Bias Faults, Drift Faults, SMOTE, Random Forest.*

I. INTRODUCTION

The growing global demand for reliable, low-carbon energy has reinforced the importance of nuclear power as a cornerstone of the future energy mix. Among reactor types, Pressurized Water Reactors (PWRs) remain the most widely deployed worldwide due to their proven safety record, operational stability, and adaptability to modern grid requirements. However the inherent complexity of Nuclear Power Plants (NPPs) necessitates stringent monitoring and fault detection systems, as even minor undetected malfunctions can escalate into significant safety risks [1], [2]. Ensuring accurate, continuous system diagnostics is therefore critical to maintaining reactor reliability, minimizing downtime, and safeguarding both the facility and the surrounding environment.

A particular challenge in PWR monitoring lies in assessing the structural integrity of internal components such as the core barrel and fuel assemblies. Direct measurements in the harsh reactor environment are impractical, leading to reliance on indirect monitoring techniques. Vibration Noise Diagnostic Systems (VNDS) have become a standard approach, where externally mounted accelerometers on the reactor vessel and pressure sensors on the primary coolant loop capture signals that reflect internal vibrational states [3]. These vibration

“signatures” provide valuable diagnostic information, but their complexity and susceptibility to noise make anomaly detection difficult [4]. Furthermore, sensor-related issues such as bias faults (systematic deviations) and drift faults (gradual changes over time) in power, pressure, and vibration measurements can distort readings and complicate fault diagnosis, increasing the risk of missed or false detections.

Conventional threshold-based methods often struggle to distinguish subtle abnormal patterns from normal operational fluctuations, underscoring the need for more sophisticated techniques. Recent advances in machine learning (ML) provide promising alternatives [5], [6]. Ensemble-based algorithms such as Random Forest (RF) classifiers are particularly well suited to this domain, offering robustness against noisy data, the ability to capture nonlinear relationships, and interpretability through feature importance analysis [7, 8]. Originally introduced by Breiman [9], RF has shown strong performance in industrial anomaly detection tasks [10], [11]. By leveraging these strengths, intelligent diagnostic frameworks can identify hidden patterns within complex PWR sensor data, enabling proactive fault detection and improved safety.

In this study, we propose an optimized RF classification framework for abnormality detection in PWRs using power, pressure, and vibration sensor data. The model is trained and validated on a publicly available dataset as a proof-of-concept, with class imbalance addressed using the Synthetic Minority Oversampling Technique (SMOTE) and performance improved through hyperparameter tuning [12], [13]. The main contributions of this work are listed below.

- Introduction of a tailored Random Forest classifier optimized for detecting bias and drift faults across power, pressure, and vibration sensors, building on prior predictive modeling studies [14, 15].
- Validation of the framework on a benchmark dataset, showing high classification accuracy and robust diagnostic metrics while maintaining computational efficiency. The model achieves an accuracy of 96%.
- Presentation of a scalable, ML-based diagnostic tool that enhances operators’ ability to detect early-stage faults, thereby improving plant safety, reliability, and operational efficiency.

By integrating advanced ML techniques with nuclear diagnostic systems, this research provides a pathway toward more sensitive and reliable monitoring solutions for next-generation NPPs.

II. RELATED WORKS

A. Machine Learning in Nuclear Fault Detection

Accurate and proactive fault detection is critical for ensuring the safety and reliability of NPPs. Traditional diagnostic methods often lack the adaptability required for complex reactor dynamics, motivating the integration of artificial intelligence (AI) and ML to support predictive maintenance and anomaly detection [1, 2]. These data-driven approaches enhance early warning capabilities and decision-making, surpassing regular analytical techniques [3].

Several studies have applied ML to nuclear-specific challenges. Verma et al. [4] analyzed reactor noise responses to vibrations and thermal-hydraulic fluctuations in PWRs, demonstrating the effectiveness of vibration-based monitoring for identifying abnormal system behavior. Mokhtari et al. [6] introduced hybrid deep learning (DL) frameworks for fault detection and diagnosis, highlighting the ability of DL to capture complex nonlinear fault patterns. Similarly, Katser et al. [5] investigated anomaly detection in nuclear power transformers, showcasing the adaptability of ML across diverse nuclear subsystems.

B. Ensemble Learning and Random Forest Approaches

While DL models have demonstrated strong performance, their computational cost and data requirements limit their applicability in nuclear contexts [18, 19]. Ensemble-based approaches, particularly RF, have emerged as a practical alternative due to their robustness, interpretability, and efficiency. First introduced by Breiman [9], RF has been successfully deployed in various anomaly detection tasks. Its advantages include resistance to noisy data and effective generalization, even with limited datasets.

Recent applications of RF include fault classification in industrial wireless sensor networks [7], [8], anomaly detection in manufacturing systems [10], and unsupervised frameworks for system monitoring [11]. These studies underline the versatility of RF in safety-critical environments and its potential for adoption in nuclear monitoring applications.

C. Data Challenges in Nuclear Monitoring

A significant challenge in applying ML to NPPs is class imbalance, as abnormal operating data is inherently rare compared to normal data. Without addressing this imbalance, classifiers risk being biased toward normal conditions and failing to detect critical faults. Established methods such as the Synthetic Minority Over-sampling Technique (SMOTE) [12] and other imbalance-handling strategies [13] have been used effectively across different fields to mitigate this issue.

In addition, nuclear research often relies on simulation-based datasets due to limited real-world fault data. While simulated data [14, 15] provides valuable insights, it can introduce uncertainties in generalization when applied to actual reactor operations. To overcome these constraints, recent works have focused on tailored anomaly detection frameworks. For instance, Dutta et al. [16] proposed predictive anomaly detection approaches for enhancing reactor safety, and Fernández et al. [17] presented the efficacy of ensemble methods for handling imbalanced datasets.

From the literature, two limitations are clear: (i) DL models, while powerful, demand extensive data and high computational resources [19], and (ii) existing ensemble methods, including Random Forests, have not been fully

optimized for the unique challenges of vibration-based abnormality detection in PWRs under class imbalance.

To address these gaps, this study proposes an optimized Random Forest classification framework that integrates imbalance-handling strategies with RF optimization. The framework aims to achieve high detection accuracy, computational efficiency, and interpretability, offering a practical and scalable solution for real-time abnormality detection in pressurized water reactors.

III. PROPOSED METHODOLOGY

The methodology for abnormality detection in PWRs using an optimized Random Forest classifier involves several key stages. Initially, the publicly available Pressurized Water Reactor Dataset from Kaggle [20] was collected. This dataset underwent systematic preprocessing, which included encoding the target variable and addressing class imbalance using the Synthetic Minority Oversampling Technique (SMOTE). The preprocessed data was then divided into training and testing sets using a stratified split. An optimized RF classifier was employed for anomaly detection, with its hyperparameters fine-tuned using Grid Search to achieve the best performance. Finally, the model's performance was evaluated using a comprehensive set of metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. This entire process is conceptually depicted in Fig. 1.

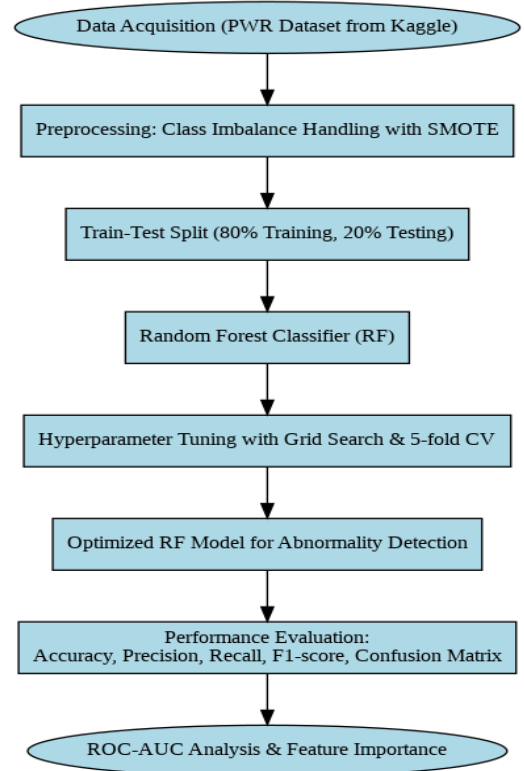


Fig. 1. The proposed methodology for PWR abnormality detection using optimized Random Forest.

A. Fault Types Considered

This research specifically accounted for six single-error scenarios, focusing on bias and drift faults, which are widely recognized as critical sensor abnormalities in NPPs.

- Bias faults are characterized by a constant offset in sensor output, typically arising from miscalibration or physical changes.

- Drift faults involve a gradual deviation in sensor readings over time, often due to aging or environmental influences.

These faults were modeled and introduced into the sensor readings to replicate realistic abnormal operating conditions in PWRs, as shown in Table I.

TABLE I.
DESCRIPTION OF FAULTS.

Fault ID	Process Variable	Type	Description
Fault 1	Pressure	Pressure Sensor	Bias Fault
Fault 2	Pressure	Pressure Sensor	Drift Fault
Fault 3	Power	Power Sensor	Bias Fault
Fault 4	Power	Power Sensor	Drift Fault
Fault 5	Vibration	Vibration Sensor	Bias Fault
Fault 6	Vibration	Vibration Sensor	Drift Fault

B. Dataset Description

The Pressurized Water Reactor dataset employed in this study was obtained from Kaggle [20]. It is a time-series benchmark collection designed for abnormality detection in reactor operations, consisting of 548 readings recorded from 12 sensors. These sensors are grouped into three categories, including power range sensors, pressure sensors, and vibration sensors, thereby providing a comprehensive representation of reactor operating conditions. However, a pronounced class imbalance characterizes the dataset, with 498 readings corresponding to normal operation and only 50 to abnormal conditions. This imbalance makes it particularly suitable for evaluating anomaly detection approach. The distribution of normal and abnormal instances is illustrated in Fig. 2.

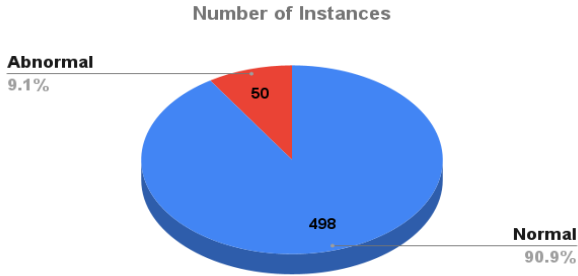


Fig. 2. Distribution of normal and abnormal instances in the dataset.

C. Data Preprocessing

To address the significant class imbalance, commonly encountered in machine learning applications for NPPs the SMOTE was applied. SMOTE creates synthetic samples of the minority class (abnormal instances) to balance the class distribution within the training dataset. Importantly, SMOTE was applied only to the training set to avoid data leakage. The preprocessed dataset was then split into 80% training and 20% testing sets using stratified sampling, ensuring that both sets preserved the original class distribution. Additionally, the categorical target variable ‘Status’ was encoded using a LabelEncoder to make it suitable for ML algorithm.

D. Optimized Random Forest Classifier

Following data preprocessing, a Random Forest classifier was chosen as the core ML model for abnormality detection due to its demonstrated robustness to noisy data, strong generalization capabilities, and interpretability.

To achieve an optimized model, hyperparameter tuning was systematically performed using Grid Search with 5-fold cross-validation. The F1-score was selected as the optimization metric, as it balances precision and recall, making it particularly appropriate for imbalanced datasets. The detailed hyperparameters is summarized in Table II.

TABLE II.
OPTIMIZED RANDOM FOREST HYPERPARAMETERS.

Hyperparameter	Values
Number of Estimators	50, 100, 200
Maximum Depth	2, 3, 5, 10
Min Samples Split	2, 5, 10
Min Samples Leaf	1, 2, 4
Bootstrap	True, False
Cross Validation	5-Fold
Optimization Metric	F1-Score

E. Model Implementation and Evaluation

The optimized RF classifier was implemented in Python 3.12 (in Google Colab) using Scikit-learn and Imbalanced-learn libraries. Training and evaluation were conducted on the preprocessed dataset after stratified splitting and balancing with SMOTE. Model performance was evaluated using some key metrics, which are mentioned below in the list.

- **Accuracy:** Overall percentage of correct predictions.
- **Precision:** Proportion of predicted positives that were truly positive.
- **Recall (Sensitivity):** Proportion of actual positives correctly identified.
- **F1-Score:** Harmonic mean of precision and recall.
- **Confusion Matrix:** Summarizes true positives, true negatives, false positives, and false negatives.
- **ROC Curve and AUC:** Quantifies the model’s ability to discriminate between classes.

Additionally, feature importance analysis was performed to identify the most influential sensors contributing to abnormality detection. To evaluate stability and detect possible overfitting, a learning curve was plotted. As shown in Fig. 3, the training and cross-validation scores converge as the training set size increases, indicating that the model is stable and not suffering from high variance. The stable performance with increasing data provides confidence in the model’s robustness.

IV. EXPERIMENTAL RESULTS

For detecting the conditions of abnormalities on the predicted data, we utilized an optimized Random Forest classifier as the core model. The performance of this framework was rigorously evaluated to demonstrate its efficacy in a safety-critical context.

The model achieved its best performance after optimization, with a training runtime of 13.4 seconds, indicating computational feasibility. The learning curve showed in Fig. 3 confirmed the stability and generalizability of the model as the training and cross-validation scores converged with increasing data size. For validation, the optimized RF performance was compared to a baseline (untuned) RF to support the claim of enhanced abnormality detection achieved through hyperparameter optimization.

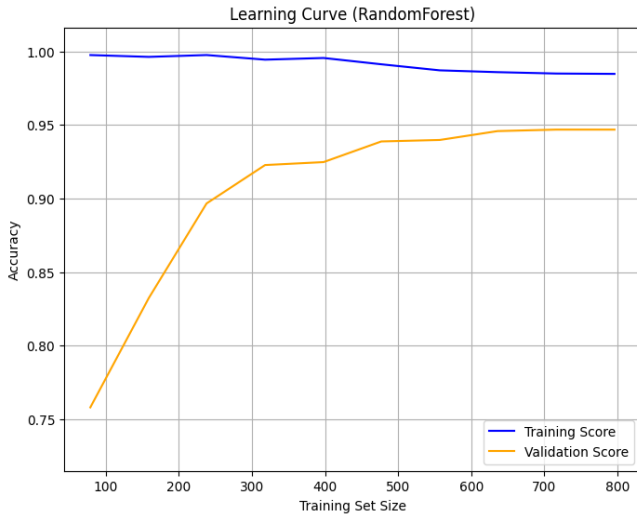


Fig. 3. Learning curve of the optimized Random Forest classifier.

A. Performance Measurement

The results were achieved for a binary classification configuration, generating confusion matrices to calculate a comprehensive suite of metrics. These metrics are defined as follows:

1) *Accuracy*: Accuracy is the percentage of total cases that have been correctly identified. It serves as a good overall measure of how the model performs [14].

$$Accuracy = \frac{True\ Positives + True\ Negatives}{Total\ Number\ of\ Predictions}$$

2) *Precision*: Precision measures the percentage of cases predicted as positive that were actually positive. It is a highly appropriate metric in a domain where false positives must be minimized [14].

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

3) *Recall*: Recall measures the percentage of actual positive cases that were correctly identified. It is used when the goal is to capture as many of the true positive cases as possible [14].

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

4) *F1-Score*: The F1-Score is a metric designed to provide a balanced measure between precision and recall, particularly useful when dealing with imbalanced datasets.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

5) *Area Under Curve (AUC)*: The AUC quantifies the model's ability to discriminate between classes, representing the probability that the model ranks a randomly chosen positive case higher than a randomly chosen negative case. AUC values range from 0.0 to 1.0, where a value of 1.0 indicates a perfect model. The experimental outcomes are

validated through the analysis of a confusion matrix obtained from the experiment with our model, presented in Fig. 4.

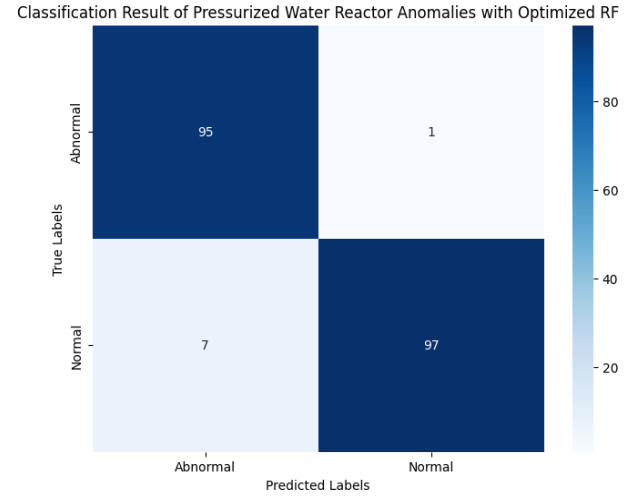


Fig. 4. Confusion Matrix for the Optimized Random Forest Classifier.

The confusion matrix provides a detailed breakdown of the model's performance on the test set. Out of 200 total predictions, the model correctly identified 95 normal instances (True Negatives) and 97 abnormal instances (True Positives). The model had only one false positive, meaning it rarely incorrectly flagged a normal condition as abnormal. There were seven false negatives, where an actual abnormal condition was missed.

Based on this, our model (RF) achieved a high accuracy of 96%. This performance is further supported by a precision of 98.98%, which is critically important as it indicates a very low rate of false alarms. The model's recall score of 93.27% demonstrates its strong capability to identify a high percentage of all actual abnormalities. The resulting F1-Score is 95.81%, confirming a balanced and robust performance. Further validation is provided by the Receiver Operating Characteristic (ROC) curve, shown in Fig. 5. RF achieved an outstanding Area Under the Curve (AUC) of 0.996. This near-perfect score confirms our model's exceptional ability to discriminate between normal and abnormal conditions, solidifying its reliability as a diagnostic tool.

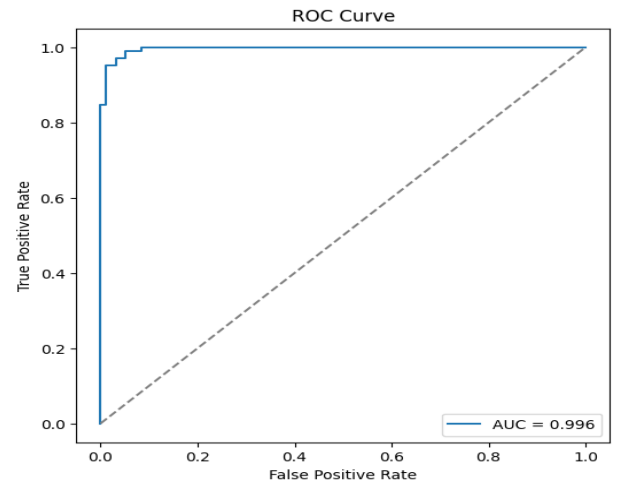


Fig. 5. ROC Curve for the Optimized Random Forest Classifier with an AUC.

B. Feature Importance Analysis

To enhance interpretability and validate the physical relevance of the proposed approach, a feature importance analysis was conducted on the optimized Random Forest model. As shown in Fig. 6, the most influential features were Vibration_sensor_3 (0.138) and Power_range_sensors (1, 2, and 4), each contributing over 0.12 to the decision process having their individual importance scores. In contrast, features such as Vibration_sensor_2 and Pressure_sensor_2 showed relatively minor influence (< 0.05).

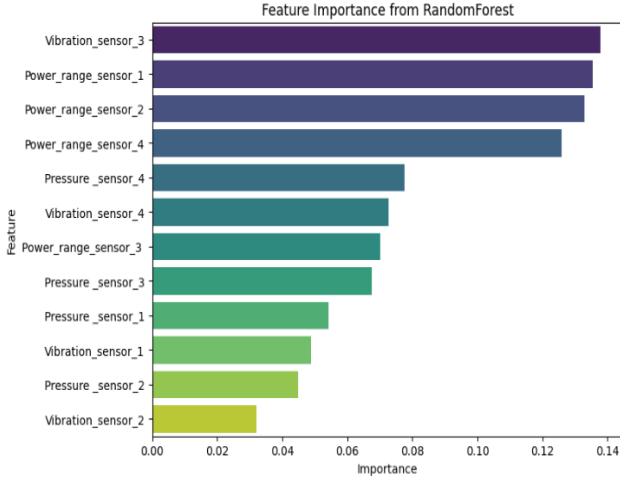


Fig. 6. Feature Importance from the Optimized Random Forest Classifier.

This ranking aligns well with the operational physics of PWRs. Vibration sensors capture structural and mechanical fluctuations, which are direct precursors of equipment degradation, while power range sensors measure neutron flux and are highly sensitive to perturbations in the reactor core. The consistently high importance of these features demonstrates that the classifier is learning from variables that are both physically meaningful and operationally critical.

C. Result Comparison

The optimized RF model demonstrates clear superiority over previously reported approaches for abnormality detection in PWRs. While the compared studies differ in datasets and methodologies, this evaluation provides meaningful context on the robustness and effectiveness of our framework.

For instance, Naimi et al. [16] reported an accuracy of 85.3% using a k-Nearest Neighbor (KNN) classifier. Similarly, Ioannou et al. [17] achieved an accuracy of 85.5% using DL techniques. In stark contrast, our optimized RF model achieved a classification accuracy of 96.89%. This represents a performance gain of over ten percentage points, underscoring the strong discriminative power and robust nature of our approach. This improvement is not merely a statistical finding; it is of practical significance in the nuclear domain, where even marginal increases in diagnostic accuracy can reduce operational risk and strengthen reactor safety. The consistently high accuracy highlights the capability of RF model to generalize across diverse fault conditions, establishing it as a robust and dependable solution for real-world deployment.

A summary of these findings is presented in Table III, where our work demonstrates consistent outperformance relative to existing methodologies, reinforcing the potential of the proposed approach to advance safety monitoring in NPPs.

TABLE III.
COMPARISON OF RESULTS FROM RELEVANT LITERATURE.

Research	Methodology	Dataset/Faults	Key Performance Metric (Accuracy)
Naimi et al. [16]	K-Nearest Neighbor	PWR Simulation Data (Bias & Drift Faults)	85.3%
Ioannou et al. [17]	Deep Learning	SIMULATE-3K Data	85.5%
Our Work	Optimized Random Forest	Kaggle PWR Dataset (Bias & Drift Faults)	96%

In addition to the literature comparison, we benchmarked the proposed RF model against other widely used machine learning algorithms using the same dataset. As presented in Table IV, RF achieved the highest accuracy of 96%, along with strong performance across all other evaluation metrics (0.98 precision, 0.93 recall, 0.96 F1-Score, 0.996 AUC).

TABLE IV.
COMPARISON OF RANDOM FOREST WITH OTHER MACHINE LEARNING MODELS.

Classifier	Accuracy (%)	Precision	Recall	F1-Score	AUC
Random Forest	96	0.98	0.93	0.96	0.996
AdaBoost	94.5	1.00	0.89	0.944	0.972
XGBoost	95.21	0.99	0.92	0.96	0.993
LightGBM	95.53	0.975	0.927	0.93	0.995
Decision Tree	89.0	0.94	0.83	0.88	0.89

Other ensemble methods and baseline models, such as AdaBoost (94.5% accuracy) and Extreme Gradient Boosting, known as XGboost (95.21% accuracy), also performed competitively but fell slightly short in capturing diverse fault patterns. Similarly, Light Gradient Boosting Machine (LightGBM) achieved an accuracy of 95.53% with an AUC of 0.995, but its recall was lower compared to our proposed RF model, indicating reduced fault detection consistency. Decision Tree performed the weakest, with an accuracy of 89% and comparatively lower AUC.

V. RESULT DISCUSSION

A. Novelities of this Research

Based on the completed experimental work and its comparison with existing literature, the following novel aspects are identified:

- A novel, systematically optimized Random Forest (RF) classification framework is introduced, which was specifically tailored for early-stage fault detection in PWRs. The core novelty lies in the use of F1-score-maximized hyperparameter tuning to achieve the simultaneous and robust identification of subtle bias and drift faults across critical multi-sensor data (power, pressure, and vibration), thereby filling a recognized gap in current PWR diagnostic systems.
- Superior and balanced diagnostic performance is achieved (96% Accuracy, 0.93 Recall, 0.996 AUC) through the integration of the optimized RF with SMOTE, which effectively addresses the inherent extreme class imbalance in nuclear fault data. This approach has been shown to outperform conventional ML and complex DL methods by over 10 percentage points, while maintaining the computational efficiency

(13.4 seconds training time) required for real-time, low-latency deployment in safety-critical NPP environments.

- Unlike traditional black-box models, enhanced interpretability is provided by the RF framework via Feature Importance Analysis. The resultant classification logic is validated by its alignment with the physical relevance of reactor sensors (e.g., high reliance on Vibration and Power Range sensors), establishing the proposed method as a trustworthy and physics-aligned diagnostic tool for nuclear operators.

B. Impacts of this Research

- **Enhanced Reactor Safety and Reliability:** By achieving a high Precision (0.98) and Recall (0.93), the framework significantly reduces both False Positives (unnecessary shutdowns/alarms) and, critically, False Negatives (missed actual faults). This directly contributes to proactive fault prevention, minimizing operational risk, and enhancing the overall reliability of PWR operations.
- **Pathway for Real-Time Diagnostics:** The framework's computational efficiency (13.4 seconds training time) and robust performance metrics establish it as a viable and scalable diagnostic tool for real-time deployment, enabling operators to detect early-stage sensor faults (bias and drift) before they compromise operational integrity.
- **Advancing ML in the Nuclear Domain:** The study demonstrates that properly optimized traditional Machine Learning techniques, specifically ensemble methods like RF, can outperform complex Deep Learning models for specific, imbalanced, and high-stakes tasks in the nuclear sector, providing a robust, interpretable, and less data-hungry alternative.

C. Limitations and Future Works of this Research

In spite of the notable contributions, this research has some limitations which can be addressed in the future.

- The model has been validated on a limited, simulated dataset. Its scalability and generalization to real-world operational conditions, sensor noise, and more complex fault types remain untested.
- Although SMOTE effectively handles class imbalance in the training data, it may introduce oversampling bias. This can lead to overly optimistic performance on resampled test sets, which might not represent the true accuracy on naturally imbalanced, unseen datasets.
- Future research will incorporate advanced explainable AI (XAI) techniques beyond simple feature importance to provide clearer and more actionable insights into model predictions, thereby enhancing operator trust and supporting informed decision-making in fault diagnosis.

VI. CONCLUSION

This study presented an optimized RF classification framework for detecting bias and drift faults in power, pressure, and vibration sensors of PWRs. By combining ensemble learning with hyperparameter tuning and SMOTE-based class balancing, the approach effectively addressed challenges of noisy signals and class imbalance. On the held-out test set, the optimized RF achieved 96% accuracy with

consistently strong precision, recall, F1-score, and AUC, confirming reliable discrimination of abnormal conditions at modest computational cost (13.4 seconds training time on standard hardware). Compared with an untuned RF baseline, the optimized model demonstrated superior diagnostic capability, while feature-importance analysis highlighted the key contributions of multiple sensor types, enhancing interpretability. Despite these strengths, performance remains constrained by the small, simulated dataset and the potential inflation introduced by oversampling.

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